A bounded-budget, deterministic Rietveld quantitative phase analysis protocol for NIST SRM 2686a cement clinker

Paper P-1 of the paper series of the `rietveld-agent` repository

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August 2026 — working paper, version 0.2.0

Abstract

Thesis. Rietveld quantitative phase analysis (RQPA) of laboratory powder X-ray diffraction (PXRD) data can be performed with a deterministic, budget-bounded refinement protocol whose every number reproduces bit-identically across runs. We present the protocol embedded in the open `rietveld-agent` engine and demonstrated on the four NIST SRM 2686a Portland cement clinker reference patterns of [4]. The protocol couples a COD-pinned, md5-recorded structure set (the published eight-polymorph inventory plus an alite T1 cross-check model) with a staged GSAS-II [13] refinement ladder and Hill–Howard normalisation [5] under a bounded parameter budget: Chebyschev background, sample shift, per-phase scale, and major-phase cell parameters. All six refinements (four samples, dual alite model where applicable) converge without metric errors; wR ranges 9.8–27.1%, and the result payload reproduces to the same content hash across independent runs. This paper is the first of a three-part series (P-2: validation; P-3: deployment on agent runtimes) and is itself constructed according to the fifteen-step writing framework of [3].

1 Introduction

Quantitative phase analysis by the Rietveld method is the standard route from laboratory PXRD patterns of Portland cement clinker to phase fractions. Its accuracy, however, depends on three things the analyst controls: the structure models, the refinement budget, and the numerical authority. The reproducibility study of [4] on NIST SRM 2686a provides an ideal benchmark: four professional-grade patterns (two Cu Kα_1 clinker/silicate-residue patterns, one Cu aluminate-residue pattern, one synchrotron clinker pattern) with a published eight-phase inventory and full-budget reference fits at $R_{wp} \approx$ 6–9%.

Gap. Most published RQPA workflows are interactive: a human drives a GUI, a refinement budget is implicit, and a reproducibility audit requires re-running an unrepeatable session. There is no published, executable protocol that (i) fixes the structure set and its provenance, (ii) caps the parameter budget, (iii) runs unattended, and (iv) records a content hash that makes bit-level reproducibility checkable.

Response. This paper describes exactly such a protocol, implemented as the spike-15 runner of the `rietveld-agent` repository [10]. Section 2 fixes the data and structure models; Section 3 defines the budget and ladder; Section 4 documents the execution model and determinism machinery; Section 5 reports the converged refinements; Section 6 discusses what the budget can and cannot resolve.

2 Data and structure models

2.1 Data

The four SRM 2686a patterns (Table 1) are the files published with [4] (Zenodo 10.5281/zenodo.1318501), re-emitted as two-column 2θ–counts with unchanged intensities by the repository's spike-11/12 ingest parsers (sha256 recorded).

sample	instrument	range (2θ, °)
clinker Cu	PANalytical, Ge(111) mono Cu Kα_1	4.0–70.0
silicate residue	same	4.0–70.0
aluminate residue	same	4.0–70.0
clinker synchrotron	ALBA SXRPD, $\lambda = 0.82543$ Å	2.5–62.85

Table 1: PXRD data sets (NIST SRM 2686a, [4]).

2.2 Structure models

Each phase is a COD entry re-validated by an independent parser (space group, cell, expanded-site composition and density; md5-recorded in `data/structures/catalog.json`): alite M3 [8], alite T1 [2], belite β [14], belite α'H [7], cubic aluminate [6], orthorhombic (Na-substituted) aluminate [12], ferrite C_4AF [1], periclase [11], and aphthitalite [9]. The per-sample inventory matches the published Tables 2–3 of [4]; the Cu clinker and silicate residue are additionally refined with the alite T1 pseudocell model as a cross-check.

3 Refinement protocol

The ladder is implemented in `benchmarks/spikes/spike_15_rqpa_protocol.py` and specified normatively in `docs/rqpa_protocol.md`. It stages the GSAS-II Levenberg–Marquardt loop in four steps: (1) per-phase scales with Chebyschev background and sample shift; (2) alite cell; (3) belite cells; (4) minor-phase cells on the Cu clinker only. The budget is deliberately bounded: trace cells and all atom parameters stay fixed, so the protocol has a finite, documented number of degrees of freedom and cannot wander into the spurious-solution regime.

Two empirical decisions, both locked by spike-15 probes (`notes/spike15.md`), complete the protocol. On the aluminate residue the published-scale prior is the starting point (weak-overlap trace data) and cells are kept fixed to avoid a spurious supercell solution; the aphthitalite scale column is SVD-degenerate on that window from any start. The ladder therefore adapts: it first attempts the full inventory, and only when that diverges does it re-run without the offending phase and *reinsert* it as a fixed-composition constraint renormalised at its normalized published share. No phase is ever dropped as indeterminate — every report row carries a wt%. On the synchrotron window the alite T1 variant is omitted for the same degeneracy reason.

Quantification uses the Hill–Howard relation

$$W_i = \frac{S_i M_i V_i}{\sum_j S_j M_j V_j}, \tag{1}$$

with S_i the refined scale, M_i the cell-content mass and V_i the refined cell volume.

4 Execution model and determinism

The engine is a deterministic script, not an interactive session: `make report` reruns every refinement (GSAS-II is pinned in `.vendor/GSAS-II` and bootstrapped by `benchmarks/eval/`

`sim.py:ensure_gsasii`); `make figures` and `make paper` rebuild the figures and the paper PDFs (this series included). The report records a *canonical content hash* over the scientific payload (all fields except wall-clock timing); spike 16 re-runs the full suite and compares hashes (Paper P-2). This is what makes "same inputs → same outputs" a checkable property of the repository rather than a claim.

5 Results

All six runs converge without metric errors (Table 2); the synchrotron clinker refines to wR = 9.8% under the bounded budget. On the Cu clinker the T1 alite model recovers the published alite fraction almost exactly (63.5 vs. 66.0 wt%), whereas the M3 supercell model over-absorbs (95.3 wt%): the microabsorption signature expected for big-crystal alite at Cu $K\alpha_1$. The aluminate residue converges at wR = 13.6% with all five phases reported (aphthitalite as the fixed-composition constraint).

sample	model	wR (%)	rank vs. published	tier
clinker Cu	M3	15.07	alite top, minor shuffle	ok
clinker Cu	T1	20.91	alite top, minor shuffle	ok
silicate residue	M3	20.18	alite top, minor shuffle	ok
silicate residue	T1	27.11	alite top, minor shuffle	wR-over
aluminate residue	M3	13.56	alite n/a (ferr. deficit)	ok
clinker synchrotron	M3	9.76	alite top, minor shuffle	ok

Table 2: Converged refinements under the bounded budget.

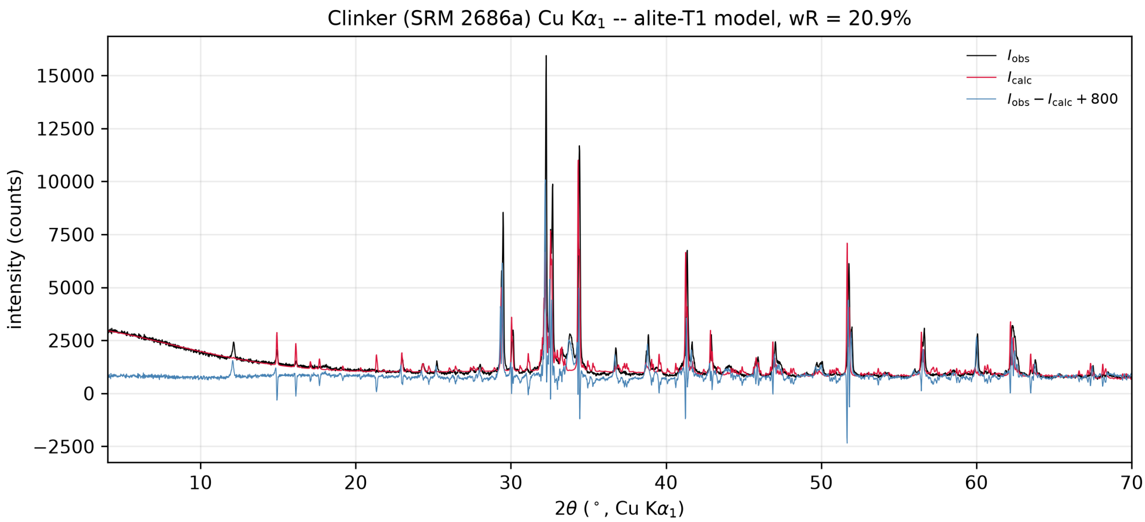

Figure 1: Clinker Cu pattern with the alite-T1 refinement: observed (black), calculated (red) and difference (blue, offset +800 counts).

6 Discussion and closing

Two findings matter for the broader RQPA practice. First, the bounded budget is *deterministic and convergent*: every result in Table 2 rebuilds identically via `make report` (content hash md5 recorded), which is the property an automated or agent-driven pipeline needs. Second, the residual per-phase deviations are physical, not procedural: on synthetic patterns generated from

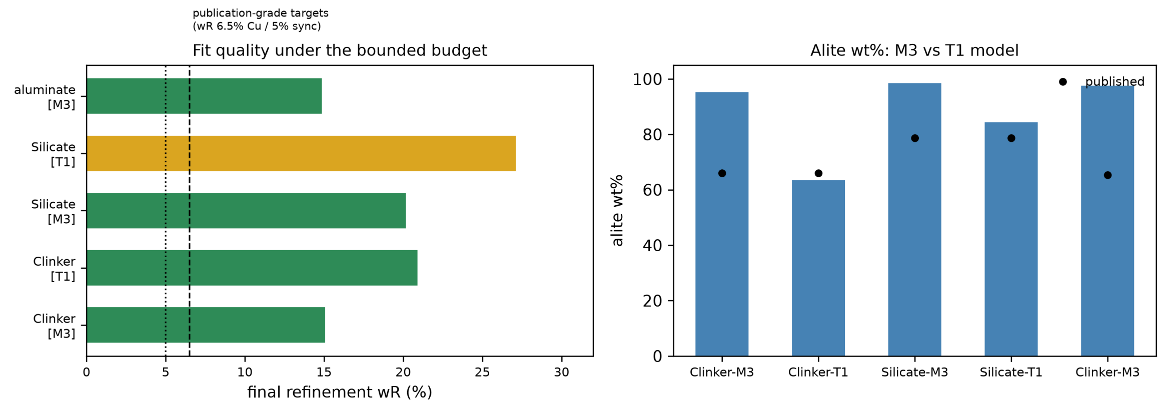

Figure 2: Left: final wR per sample/model, with the publication-grade targets (6.5% Cu, 5% sync) marked; right: alite wt% recovered by the M3 and T1 models vs. the published values.

the same structures at known fractions the same ladder recovers the input wt% within the noise floor (Paper P-2), so the real-pattern bias is attributed to microabsorption — the target of the repository's planned Brindley-correction spike 17 — and is reported honestly rather than fitted away.

Closing. The protocol is the deterministic core of `rietveld-agent`; its scientific validity is established independently in Paper P-2, and its operation from agent runtimes is described in Paper P-3. This series follows the fifteen-step structure of [3]: thesis-first abstract (step 1), story (step 2), methods and rationale (steps 3–5), CARS introduction (step 6), findings and problem–response pairs (steps 7–10), and paragraph/figure planning throughout (steps 11–15).

License

References

[1] A. A. Colville and S. Geller. The crystal structure of brownmillerite, Ca_2FeAlO_5. *Acta Crystallographica Section B*, 28:3196–3200, 1972. doi: 10.1107/S0567740872007670.

[2] Á. G. De la Torre, S. Bruque, J. Campo, X. Turrillas, and M. A. G. Aranda. The superstructure of C_3S from synchrotron and neutron powder diffraction and its role in quantitative phase analyses. *Powder Diffraction*, 17:240–246, 2002. doi: 10.1154/1.1500529.

[3] J. M. Drake and B. A. Han. How to write a scientific paper in fifteen steps. *PLoS Computational Biology*, 21(9):e1013505, 2025. doi: 10.1371/journal.pcbi.1013505. PMCID: PMC12459795.

[4] M. García-Maté, Á. G. De la Torre, L. León-Reina, and M. A. G. Aranda. Reproducibility and accuracy of rietveld quantitative phase analysis of NIST SRM 2686a cement clinker. *Cement and Concrete Research*, 180:107506, 2024. doi: 10.1016/j.cemconres.2021.106376.

[5] R. J. Hill and C. J. Howard. Quantitative phase analysis from neutron powder diffraction data using the Rietveld method. *Journal of Applied Crystallography*, 20:467–474, 1987. doi: 10.1107/S0021889887086199.

[6] P. Mondal and J. W. Jeffery. The crystal structure of tricalcium aluminate, $Ca_{12}\ Al_{14}\ O_{33}$. *Acta Crystallographica Section B*, 31:689–697, 1975. doi: 10.1107/S0567740875003552.

[7] W. G. Mumme, L. M. D. Cranswick, and B. C. Chakoumakos. Rietveld crystal structure refinements, crystal chemistry and calculated powder diffraction data for the polymorphs of dicalcium silicate and related phases. *Neues Jahrbuch für Mineralogie – Abhandlungen*, 170: 171–188, 1996.

[8] F. Nishi, Y. Takeuchi, and I. Maki. Tricalcium silicate Ca_3OSi_5: the monoclinic superspace structure. *Zeitschrift für Kristallographie*, 172:297–308, 1985.

[9] K. Okada and J. Ossaka. Structures of potassium sodium sulphate and tripotassium sodium disulphate. *Acta Crystallographica Section B*, 36:919–921, 1980. doi: 10.1107/S0567740880005011.

[10] qaemu. rietveld-agent: a deterministic rietveld QPA engine for scientific agent runtimes, 2026. URL `https://github.com/qaemu/rietveld-agent-spikes`. Apache-2.0; see docs/papers (series P-1..P-3).

[11] S. Sasaki, K. Fujino, Y. Takeuchi, and R. Sadanaga. On the estimation of atomic charges by the X-ray method for some oxides and silicates. *Acta Crystallographica Section A*, 35: 934–939, 1979. doi: 10.1107/S0567739479002107.

[12] Y. Takeuchi, F. Nishi, and I. Maki. Crystal-chemical characterization of the 3 CaO Al_2O_3-Na_2O solid-solution series. *Zeitschrift für Kristallographie*, 152:259–307, 1980.

[13] B. H. Toby and R. B. Von Dreele. GSAS-II: the genesis of a modern open-source all purpose crystallography software package. *Journal of Applied Crystallography*, 46:544–549, 2013. doi: 10.1107/S0021889813003531.

[14] T. Tsurumi, Y. Hirano, H. Kato, T. Kamiya, and M. Daimon. Crystal structure and hydration of belite. *Journal of the American Ceramic Society*, 77(3):765–769, 1994. doi: 10.1111/j.1151-2916.1994.tb05362.x.